Simpler Model Link Seed and Growth Directly

Data

For observations $n = 1, \ldots, N$:

- $sc_n \in \mathbb{N}_0$: seed count
- $rw_n \geq 0$: radial growth
- $DBH_n \geq 0$: DBH
- $GST_n \in \mathbb{R}$: growing season temperature

Parameters

$$\alpha \in \mathbb{R}$$

$$\beta_{\text{rw}} \in \mathbb{R}$$

$$\beta_{\text{dbh}} \in \mathbb{R}$$

$$\beta_{\text{GST}} \in \mathbb{R}$$

Prior Distributions

$$\alpha \sim \mathcal{N}(0, 5)$$

$$\beta_{\text{rw}} \sim \mathcal{N}(0, 5)$$

$$\beta_{\text{dbh}} \sim \mathcal{N}(0, 5)$$

$$\beta_{\text{GST}} \sim \mathcal{N}(0, 5)$$

$$\phi_{sc} \sim \text{Gamma}(2, 0.1)$$

Model

For each observation n:
Expected seed count:

$$\log(\mu_{sc,n}) = \alpha + \beta_{rw} \cdot rw_n + \beta_{dbh} \cdot DBH_n + \beta_{GST} \cdot GST_n$$

Observation:

$$sc_n \sim \text{NegBinomial}\left(\log(\mu_{sc,n}), \phi_{sc}\right)$$